

Day: _____

Lab Assignment 1

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Course:

Applied physics Lab

Submitted to:

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Lab Assignment 1

Question No 1

Implement a series circuit consisting of three resistors $R_1 = 50\Omega$, $R_2 = 60\Omega$ and $R_3 = 30\Omega$ connected across on 19V DC supply. Calculate the total current flowing in the circuit using ohm's Law, and construct the circuit for 9 resistors.

Given Data:

$$R_1 = 50\Omega$$

$$R_2 = 60\Omega$$

$$R_3 = 30\Omega$$

$$V = 19V$$

To Find:

$$I_{\text{total}} = ?$$

Construct the circuit for 9 resistors. (In series)

Series Circuit: In series circuit, components are connected end-to-end in a single loop. Because there is only one path for the electrons flow, the current (I) remains constant throughout the entire circuit. This means total current is equal to the current flowing through each individual resistor ($I_{\text{total}} = I_1 = I_2 = I_3$)

Total Resistance:

The total resistance is sum of all individual resistances.

$$R_{\text{total}} = R_1 + R_2 + R_3 + \dots + R_n$$

Formula to calculate total current using ohm's law;

$$I_{\text{total}} = \frac{V}{R_{\text{total}}}$$

First we find total (or equivalent) resistance,

$$R_{\text{total}} = R_1 + R_2 + R_3$$

Put given values in this formula,

$$R_{\text{total}} = 50\Omega + 60\Omega + 30\Omega$$

$$R_{\text{total}} = 140\Omega$$

Total current:

Find total current by using ohm's law,

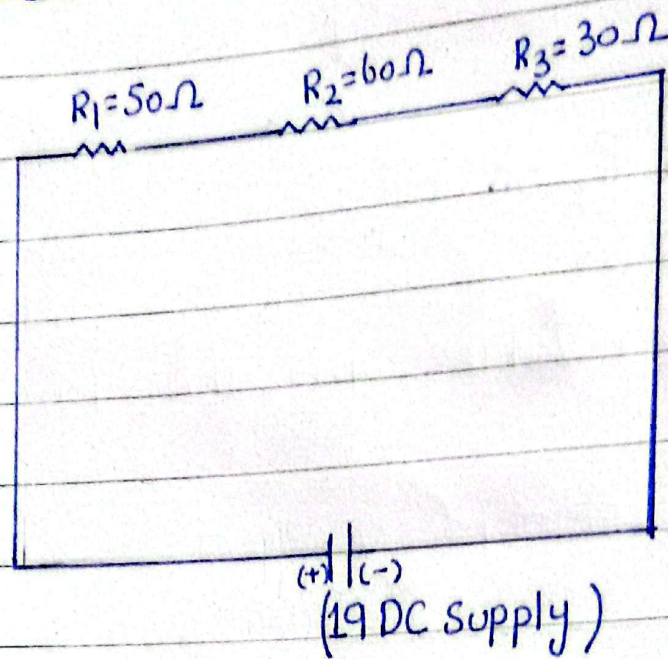
$$I_{\text{total}} = \frac{V}{R_{\text{total}}}$$

Put values in this formula,

$$I_{\text{total}} = \frac{19V}{140\Omega}$$

$$I_{\text{total}} = 0.1357 A$$

Series circuit:



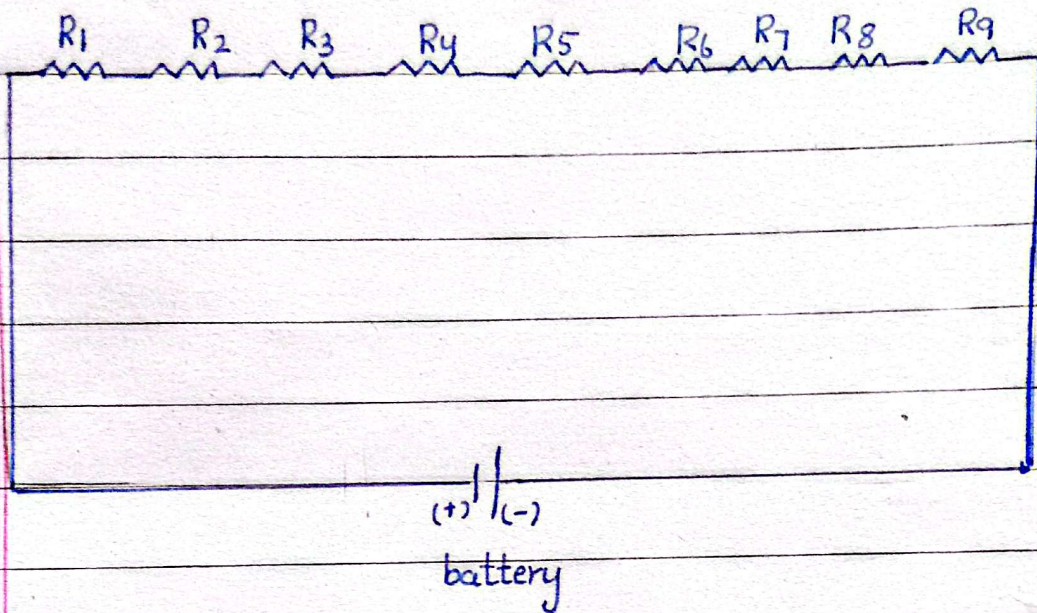
Construct Circuit for 9 Resistors: (Series Circuit)

Construction Principle:

To construct a series circuit with 9 resistors, connect the resistance end-to-end so that there is only path for electrical flow. The tail of R_1 connects to the head of R_2 and R_2 tail connect to head of R_3 , all the

way to R_9 . R_1 and R_9 also connect battery terminals.

Construction:



In this circuit, if we find

$$R_{\text{total}} = R_1 + R_2 + R_3 + R_4 + R_5 + R_6 + R_7 + R_8 + R_9$$

and by using ohm's law,

$$I_{\text{total}} = \frac{V}{R_{\text{total}}}$$

Appratus:

Apparatus to construct this in lab is nine resistors, breadboard, two connecting wires and multimeter.